

Arithmetic determinant transport in the original tau-based cohomology

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This calculation retains the arithmetic coefficient complex over the absolute pointed base $\mathbf{F}_{1,\tau} = \{\tau, 1\}$. It computes the signed arithmetic/reference determinant term on the original finite theta source. The new formulas are a complete first and second variation, an exact four-window projection trace, and a quantitative bound in terms of the actual common source Gram. No arithmetic source multiplier is assumed bounded above or bounded away from zero.

1 Original objects and the fixed comparison

The retained tau-base construction has the four-point chart $+, - < \eta < \sigma$ and the coefficient sheaf $\mathcal{T}$ with $(\mathcal{T}_+, \mathcal{T}_-, \mathcal{T}_\eta, \mathcal{T}_\sigma) = (V, V, \mathcal{B}, 0)$. Its two nonzero restrictions are Θ and $J\Theta = \Theta\mathcal{F}$. The actual projective resolution gives

$$\mathbf{A}_\tau(\mathcal{T}) = [V \oplus V \xrightarrow{\Theta(\phi-\hat{\psi})} \mathcal{B}], \quad H^1\mathbf{A}_\tau(\mathcal{T}) = Q = \mathcal{B}/\Theta V. \quad (\text{AT1})$$

These are the fixed source objects of the original Tau Base note, equations (18)–(25), supplied in the source closure. The tensor resolution in its equations (46)–(48) retains the original differential signs and identifies its top cohomology with $Q^{\otimes k}$. The operation here uses that pushforward, with its arithmetic classes. Its separately computed restriction to σ remains an available comparison; it is not substituted for (AT1).

Fix an original actual packet h of full zero orders of $g = 2\xi$, and a positive integer k . Retain the full tensor-sum operator, its original cyclic annihilator $\chi = \chi_{h,k}$ of degree $q \geq 1$, and $E = \mathbb{C}[S]/(\chi)$ in the increasing-power remainder frame. The original construction supplies

$$\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad J^{(k)}\mathcal{V}P = \eta_{h,k}J_N P, \quad \eta_{h,k}[P] = v_h^{\otimes k}P(A_h^{(k)})1, \quad (\text{AT2})$$

where $v_h = j_h(g/h)$ is the full invertible Taylor unit, J_N is literal remainder modulo χ , and $\eta_{h,k} : E \hookrightarrow E_h^{\otimes k}$ is the original injective full-jet observation, $E_h = \mathbb{C}[s]/(h)$. The original section $\sigma_h : E_h \hookrightarrow Q$ gives the separate arithmetic cohomology injection and equality

$$q^{(k)}\mathcal{V}P = \sigma_h^{\otimes k}\eta_{h,k}J_N P, \quad \sigma_h^{\otimes k}\eta_{h,k} : E \hookrightarrow Q^{\otimes k}. \quad (\text{AT2a})$$

The supplied arithmetic-input and cyclic-sum proofs construct these maps; all operations below keep them fixed. In particular the sum coordinate is $S = k/2 + iu$, with no division by k .

For $N \geq q - 1$, set $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$, $\mathcal{P}_{-1} = 0$, and let $B_N : \mathcal{P}_{N-q} \rightarrow \mathcal{P}_N$ be multiplication by the literal monic χ . Write $s_N : E \rightarrow \mathcal{P}_N$ for the degree-less-than- q remainder section. Polynomial division proves

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{B_N} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0, \quad J_N s_N = I_E. \quad (\text{AT3})$$

The arithmetic norm has density $m_1(u) = m_{h,k}(u) = w_h^{*k}(u)$, $w_h(u) = |(g/h)(1/2 + iu)|^2/(2\pi)$. The retained Gamma reference has density

$$m_0(u) = r_{1/4,k}(u) = \frac{(\sqrt{2\pi})^k |\Gamma(k/4 + iu/2)|^2}{c_{k/4} 2\pi}, \quad c_a = 2^{1-2a}\Gamma(2a). \quad (\text{AT4})$$

Its mass is $(2\pi)^{k/2}$; the arithmetic mass is the original μ_h^k . The source proofs included with this calculation establish finite moments of every degree and positivity almost everywhere for both densities. The calculation below only integrates polynomials against these two actual measures and their sum. It needs no estimate on the pointwise ratio m_1/m_0 .

For $0 \leq t \leq 1$ define, in the unchanged coefficient frame,

$$\begin{aligned} m_t &= (1-t)m_0 + tm_1, \quad \Delta m = m_1 - m_0, \quad v_N(u) = (1, S(u), \dots, S(u)^N), \\ M_N(t) &= \int v_N(u)^* v_N(u) m_t(u) du, \quad \Delta M_N = M_N(1) - M_N(0), \\ H_N(t) &= B_N^* M_N(t) B_N. \end{aligned} \tag{AT5}$$

Every $M_N(t)$ is positive definite: a nonzero polynomial has finitely many real zeros, and m_t is positive almost everywhere. B_N is injective, so $H_N(t)$ is positive definite on its actual relation domain. When $N = q - 1$, that domain is zero; its determinant is one, its unique endomorphism is its inverse, and all compositions through that domain are zero. The real parameter t interpolates norms on (AT3); it changes neither the absolute base nor a cochain map.

2 Canonical sections, determinant lines and full variations

Theorem 1. *On the fixed exact sequence (AT3), its least-norm section and quotient Gram are*

$$\begin{aligned} R_N(t) &= s_N - B_N H_N(t)^{-1} B_N^* M_N(t) s_N, \\ J_N R_N(t) &= I_E, \quad B_N^* M_N(t) R_N(t) = 0, \\ G_N(t) &= R_N(t)^* M_N(t) R_N(t) = (J_N M_N(t)^{-1} J_N^*)^{-1}, \\ V_N(t) &:= \det G_N(t) = \frac{\det M_N(t)}{\det H_N(t)}. \end{aligned} \tag{AT6}$$

In the original polynomial frame the exact first and second variations are

$$\begin{aligned} \dot{R}_N &= -B_N H_N^{-1} B_N^* \Delta M_N R_N, \quad \dot{G}_N = R_N^* \Delta M_N R_N, \\ \ddot{G}_N &= -2R_N^* \Delta M_N B_N H_N^{-1} B_N^* \Delta M_N R_N \preceq 0. \end{aligned} \tag{AT7}$$

The exact logarithmic variation, with both trace terms retained, is

$$\begin{aligned} \frac{d}{dt} \log V_N &= \text{Tr}(M_N^{-1} \Delta M_N) - \text{Tr}(H_N^{-1} B_N^* \Delta M_N B_N) \\ &= \text{Tr}(G_N^{-1} R_N^* \Delta M_N R_N). \end{aligned} \tag{AT8}$$

Its second derivative is

$$\begin{aligned} \frac{d^2}{dt^2} \log V_N &= -2 \text{Tr}(G_N^{-1} R_N^* \Delta M_N B_N H_N^{-1} B_N^* \Delta M_N R_N) \\ &\quad - \text{Tr}(G_N^{-1} \dot{G}_N G_N^{-1} \dot{G}_N) \leq 0. \end{aligned} \tag{AT9}$$

Proof. Every lift of $x \in E$ is $s_N x + B_N y$. Expanding its squared norm and completing its finite quadratic expression gives the unique minimizer $y = -H_N^{-1} B_N^* M_N s_N$. This proves the first two lines of (AT6). For the inverse expression, every polynomial vector p decomposes as $B_N y + R_N J_N p$. Orthogonality therefore gives $R_N^* M_N p = G_N J_N p$ for every p . Taking adjoints yields $M_N R_N = J_N^* G_N$. Substituting this into $J_N R_N = I_E$ gives $(J_N M_N^{-1} J_N^*) G_N = I_E$, which proves the inverse expression.

The determinant calculation preserves orientation. Put $r = N - q + 1$ and $L_N = [B_N \ s_N]$. In the ordered monomial bases, $[s_N \ B_N]$ has determinant one, since χ is monic and the lower diagonal block is triangular with diagonal one. Thus $\det L_N = (-1)^{qr}$. The upper-left block of $L_N^* M_N L_N$ is H_N , and its Schur complement is G_N . Consequently $\det H_N \det G_N = |(-1)^{qr}|^2 \det M_N = \det M_N$. Equivalently the map

$$\begin{aligned} \det \mathcal{P}_{N-q} \otimes \det E &\longrightarrow \det \mathcal{P}_N, \\ (y_1 \wedge \dots \wedge y_r) \otimes (x_1 \wedge \dots \wedge x_q) &\longmapsto B_N y_1 \wedge \dots \wedge B_N y_r \wedge s_N x_1 \wedge \dots \wedge s_N x_q \end{aligned} \tag{AT10}$$

is the fixed determinant-line isomorphism. Replacing any lift by a relation does not change its wedge. Orthogonal lifts give the product metric just computed. This proves the determinant identity including the empty relation space, without changing any scalar mass.

Differentiate $J_N R_N = I$; then $\dot{R}_N$ lies in $\text{im } B_N$. Differentiating $B_N^* M_N R_N = 0$ determines its unique coefficient and proves the first identity of (AT7). On differentiating $R_N^* M_N R_N$, the two terms involving $\dot{R}_N$ vanish by that orthogonality. A second differentiation gives the two equal negative terms in (AT7), since ΔM_N is Hermitian. This proves the claimed second derivative and its sign. Applying the finite determinant derivative to (AT6) gives the first equality of (AT8); applying it to G_N gives the second. One more derivative gives (AT9). The first trace there is the squared Hilbert–Schmidt norm of $H_N^{-1/2} B_N^* \Delta M_N R_N G_N^{-1/2}$; the second is the squared Hilbert–Schmidt norm of $G_N^{-1/2} \dot{G}_N G_N^{-1/2}$. These identities establish their signs and retain the original forms. All inverse matrices are smooth on a neighborhood of the compact interval $[0, 1]$, by positivity and continuity. No infinite-dimensional inverse has been used. $\square$

The derivative $\dot{R}_N$ in (AT7) is a literal polynomial relation. Applying the original $\mathcal{V}$ in (AT2) therefore retains its original theta primitive and sends its arithmetic class to zero; $J^{(k)} \mathcal{V} R_N(t) = \eta_{h,k}$ for every t . At each nonempty original support label A , the cochain identities lift by $(A, P) \mapsto (A, fP)$, while $\tau \mapsto \tau$. Thus a relation-valued derivative has a supported zero class at label A . Its source vector and its full nonnegative energy in (AT7)–(AT9) remain present. These formulas compute their contribution rather than deleting it after taking cohomology.

3 The signed four-endpoint term on one retained source

Let $N_* = 2q$. Every space in (AT3) for the four endpoints sits in $\mathcal{P}_{N_*}$ by its actual coefficient inclusion. Give this common space the form $M_*(t) = M_{N_*}(t)$. Write $P_N(t)$ for its orthogonal projection onto $\mathcal{P}_N$, and $Q_N(t)$ for its orthogonal projection onto the relation subspace $\mathcal{R}_N = \chi \mathcal{P}_{N-q}$. The latter is zero at $N = q - 1$. For the coefficient inclusion $I_N : \mathcal{P}_N \hookrightarrow \mathcal{P}_{N_*}$, the formulas, with their domains retained, are

$$P_N = I_N M_N^{-1} I_N^* M_*, \quad Q_N = I_N B_N H_N^{-1} B_N^* I_N^* M_*, \quad C(t) = M_*(t)^{-1} \Delta M_*. \quad (\text{AT11})$$

Both P_N and Q_N are $M_*(t)$ -self-adjoint idempotents with their displayed images, verified by substitution. Since $\mathcal{R}_N \subset \mathcal{P}_N$, $P_N Q_N = Q_N P_N = Q_N$. Their difference is the orthogonal projection onto the finite canonical quotient section and has trace q . Cyclicity of finite matrix trace in (AT11) proves the exact common-source version of (AT8):

$$\frac{d}{dt} \log V_N(t) = \text{Tr}((P_N(t) - Q_N(t))C(t)). \quad (\text{AT12})$$

Put $i_0 = q - 1$, $i_1 = q$, $j_0 = 2q - 1$, and $j_1 = 2q$. Define the two positive operators

$$\begin{aligned} U(t) &= (Q_{j_0} - Q_{i_0}) + (Q_{j_1} - Q_{i_1}), \\ W(t) &= (P_{j_0} - P_{i_0}) + (P_{j_1} - P_{i_1}). \end{aligned} \quad (\text{AT13})$$

Each difference in this display is an orthogonal projection, because the two spaces in its pair are nested in the same Hilbert space. Each has rank q . The sums need not be projections and are not asserted to be orthogonal sums; they are positive operators with $\text{Tr } U = \text{Tr } W = 2q$.

Write

$$\begin{aligned} \mathcal{B}(t) &= \log \frac{V_{i_0}(t) V_{i_1}(t)}{V_{j_0}(t) V_{j_1}(t)}, \quad T_N = V_N(1)/V_N(0), \\ \Delta \mathcal{B} &= \mathcal{B}(1) - \mathcal{B}(0) = \log \frac{T_{q-1} T_q}{T_{2q-1} T_{2q}}. \end{aligned} \quad (\text{AT14})$$

Subtracting the four exact formulas (AT12), with their original signs, and integrating gives

$$\Delta \mathcal{B} = \int_0^1 \text{Tr}((U(t) - W(t))C(t)) dt. \quad (\text{AT15})$$

This is the actual arithmetic correction in the audit's formula (54). It compares the two added relation windows with the two added polynomial windows. Concavity of each individual $\log V_N(t)$ in (AT9) does not assign a sign to their signed combination in (AT15); that combination has now been computed.

There is also an exact pointwise integral, using no division by the arithmetic density. Define

$$\begin{aligned} k_N(t, u) &= v_N(u)M_N(t)^{-1}v_N(u)^*, \\ r_N(t, u) &= |\chi(S(u))|^2 v_{N-q}(u)H_N(t)^{-1}v_{N-q}(u)^*, \\ d_N(t, u) &= k_N(t, u) - r_N(t, u), \quad r_{q-1}(t, u) = 0, \\ d_N(t, u) &= v_N(u)R_N(t)G_N(t)^{-1}R_N(t)^*v_N(u)^* \geq 0. \end{aligned} \tag{AT16}$$

The last equality follows by writing the orthogonal splitting of M_N^{-1} as $B_N H_N^{-1} B_N^* + R_N G_N^{-1} R_N^*$; multiplying by M_N and evaluating on $\text{im } B_N \oplus \text{im } R_N$ proves that inverse identity. In particular $\int d_N(t, u)m_t(u) du = q$. Consequently (AT8) also reads

$$\frac{d}{dt} \log V_N(t) = \int d_N(t, u) \Delta m(u) du, \quad \Delta \mathcal{B} = \int_0^1 \int (d_{i_0} + d_{i_1} - d_{j_0} - d_{j_1})(t, u) \Delta m(u) du dt. \tag{AT17}$$

All coefficients of these fixed-degree polynomials in u and $\bar{S}(u)$ are bounded for $t \in [0, 1]$. Since $|\Delta m| \leq m_0 + m_1$ and these measures have the required finite moments, absolute integrability proves the exchange of integrals. This remains valid at every zero of m_1 .

4 An exact finite-source bound for the correction

Let $b_{\min}$ and $b_{\max}$ be the extreme eigenvalues of the actual positive $M_*(0)$ -self-adjoint operator $D_* = M_*(0)^{-1}M_*(1)$, and set $\kappa_* = b_{\max}/b_{\min}$. These are finite original source data at the common cutoff $N_* = 2q$. They are not presumed uniform in either the packet or k .

Theorem 2. *The exact correction on the original finite source satisfies*

$$\boxed{|\Delta \mathcal{B}| \leq 2q \log \kappa_*} \tag{AT18}$$

Equivalently, retaining the original Gamma and arithmetic endpoint quantities,

$$\mathcal{B}_{h,k}^\Gamma - 2q \log \kappa_* \leq \mathcal{B}_{h,k}^{\text{ar}} \leq \mathcal{B}_{h,k}^\Gamma + 2q \log \kappa_*. \tag{AT19}$$

Proof. The operator $C(t)$ is $M_*(t)$ -self-adjoint. If its extreme eigenvalues are $c_{\min}(t), c_{\max}(t)$, positivity of U, W and their exact traces give

$$|\text{Tr}((U - W)C)| \leq 2q(c_{\max}(t) - c_{\min}(t)). \tag{AT20}$$

For example $\text{Tr}(UC) \leq c_{\max} \text{Tr } U$ follows by taking the trace of the product of U and the positive operator $c_{\max}I - C$ in the original Hilbert space. The lower bound is identical, and subtraction proves (AT20), without assuming that U, W, C commute.

In the fixed coefficient space,

$$C(t) = ((1 - t)I + tD_*)^{-1}(D_* - I).$$

Thus each eigenvalue is $c_b(t) = (b - 1)/(1 - t + tb)$ for an eigenvalue $b > 0$ of D_* . Its derivative with respect to b is $(1 - t + tb)^{-2} > 0$. The extreme values therefore correspond to the same $b_{\min}, b_{\max}$ throughout the interval, including repeated eigenvalues. Direct integration gives

$$\int_0^1 (c_{\max}(t) - c_{\min}(t)) dt = \log b_{\max} - \log b_{\min} = \log \kappa_*. \tag{AT21}$$

Integrating (AT20) in the exact identity (AT15) proves (AT18). No pointwise multiplier bound or asymptotic estimate has entered. $\square$

When $M_*(1) = cM_*(0)$ with its literal $c > 0$, the exact source form at time t is $(1 - t + tc)M_*(0)$. Each quotient determinant is $(1 - t + tc)^q V_N(0)$, so all four factors in (AT14) cancel and $\Delta\mathcal{B} = 0$. This is also forced by $\kappa_* = 1$ in (AT18). The individual masses and volume factors remain as written; cancellation occurs only in this specified ratio. An arbitrary fixed change of the coefficient frame conjugates D_* , transports every subspace and its projection, and multiplies each quotient determinant by the same squared frame determinant. Hence (AT15)–(AT21) preserve their original values under that explicitly stated map.

For the reflection-stable quartet calculation of the supplied Gamma audit, let its actual displacement be $\delta > 0$ and retain $q = [1 + k(m - 1)](k + 1)^2$. Its fully written Gamma theorem gives $\mathcal{B}_{h,k}^\Gamma \geq 4q \log(\delta k / (2\sqrt{5}))$. Substitution in (AT19) gives the new quantitative comparison on the same source:

$$\mathcal{B}_{h,k}^{\text{ar}} \geq 4q \log\left(\frac{\delta k}{2\sqrt{5}}\right) - 2q \log \kappa_*, \quad \log \kappa_* \geq 2 \log\left(\frac{\delta k}{2\sqrt{5}}\right) - \frac{\mathcal{B}_{h,k}^{\text{ar}}}{2q}. \quad (\text{AT22})$$

The quartet and actual packet restriction are retained in this application; a reference test polynomial has not been declared to divide 2ξ . Equations (AT15) and (AT17) give the full signed quantity behind this bound. In particular they provide the exact integrand that an arithmetic cancellation calculation must evaluate, with both relation windows and both polynomial windows present. No subthreshold arithmetic bound or weight purity is asserted by this source comparison alone.

5 Source dependencies and precise status

The complete retained Tau Base note supplies (AT1), its tensor pushforward, right adjoint and supported maps. The original arithmetic-input and cyclic-sum proofs supply (AT2), full packet units and the theta primitives. The Gamma/source-density notes supply the actual measures and finite moments in (AT4)–(AT5). The new Deligne-bound note supplies the full quartet Gamma lower bound used only in (AT22). Each is preserved in the accompanying source closure, with its actual edition and hash.

The new proved calculations here are (AT6)–(AT21) on those unchanged input objects, and their substitution (AT22). They give an analytic calculation on the canonical finite complex mapping to $\mathbf{A}_{\tau,k}(\mathcal{T}^{\boxtimes k})$. They neither replace that pushforward by a generic stalk nor deduce a verdict about the entire tau-based program from an auxiliary model.